

Cloud development A

CLDV6211 ICE task 3

Narishka Sewraj

ST10433872

An overview of Azure Container Registry (ACR) is given in this study, along with information on its main features, advantages, and real-world applications using image versions. For all kinds of container deployments, Azure Container Registry enables you to create, save, and administer container images and artefacts in a private registry. Utilise Azure container registries in conjunction with your current workflows for container development and deployment (dlepov, n.d.). Private Docker container images and associated content formats, including Helm charts, OCI artefacts, and images created in accordance with the OCI image format specification, are managed by Azure Container Registry (azure.microsoft.com, n.d.).

Benefits of ACR:

- **Private Image Storage:** Images in Azure Container Registry (ACR) are private, accessible only to authorized individuals and systems, unlike public Docker repositories.
- **Geo-Replication:** Supports geo-replication to maintain a single registry replicated across multiple regions, ensuring reliable, scalable, and fast image access.
- **Azure Integration:** Seamlessly integrates with Azure Kubernetes Service (AKS) and other Azure services as part of the Azure ecosystem.
- **Automated Builds and Deploys:** ACR Tasks enable automation of Docker image builds and deployments triggered by source code or image updates.
- **Enterprise-Grade Security:** Integrates with Azure Active Directory for robust authentication and authorization, and offers security features like vulnerability detection, image signing, and content trust. (Santiago, 2023).

Use cases of ACR:

- **Automated and On-Demand Image Creation:** ACR enables building container images using automated processes or on-demand workflows.
- **Secure Image Storage:** Provides a secure environment to store container images, ensuring they are readily available for use.
- **Support for Multiple Container Formats:** Stores various container formats, including Docker images, Helm charts, and Open Container Initiative (OCI) artifacts.

- Container Image Lifecycle Management: Manages the full lifecycle of container images.
- Image Integrity and Security: Supports versioning, tagging, and security scanning to ensure image integrity.(dlepow, n.d.).

Reference list:

azure.microsoft.com. (n.d.). Azure Container Registry | Microsoft Azure. [online] Available at: <https://azure.microsoft.com/en-us/products/container-registry> [Accessed 14 May 2025].

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Santiago, M. (2023). Azure Container Registry (ACR) - Santiago - Medium. [online] Medium. Available at: <https://medium.com/@barbieri.santiago/azure-container-registry-acr-1d4dd0eaaaa3> [Accessed 14 May 2025].